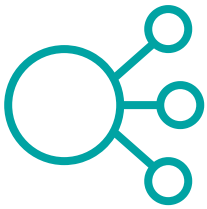

AWS ELB & Auto Scaling — Complete Notes

Load balancing + elasticity · Distribute traffic, scale automatically

A reader-friendly, all-in-one reference for Elastic Load Balancing & EC2 Auto Scaling.

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1. ⚖️ Why ELB & Auto Scaling

These two services together give you **high availability** and **elasticity**:

- **Elastic Load Balancing (ELB)** spreads incoming traffic across multiple healthy targets in multiple AZs, so no single instance is a bottleneck or a single point of failure.
- **EC2 Auto Scaling (ASG)** automatically adds/removes instances to match demand and **replaces unhealthy ones** — self-healing.

> 💡 **Mental model:** the **ELB is the front door** that splits traffic; the **ASG is the engine room** that grows/shrinks the fleet and heals it. Together: traffic always reaches a healthy instance, and capacity follows demand.

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2. 🔀 Load Balancer Types

Type	Layer	Routes on	Best for
ALB (Application)	L7 (HTTP/HTTPS)	Host, path, headers, query, method	Web apps, microservices, containers
NLB (Network)	L4 (TCP/UDP/TLS)	IP + port	Ultra-low latency, millions of req/s, static IP
GWLB (Gateway)	L3	IP packets	Inline virtual appliances (firewalls, IDS/IPS)
CLB (Classic, legacy)	L4/L7	basic	Old accounts only — avoid for new work

> Pick **ALB** for HTTP(S) apps, **NLB** for extreme performance / TCP / a static IP, **GWLB** to insert security appliances.

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3. 🌐 ALB (Application Load Balancer)

Operates at **Layer 7** — understands HTTP, so it can route on content.

- **Listener** — checks for connections on a port/protocol (e.g. HTTPS:443) and applies **rules**.

- **Rules** — route by **host** (`api.example.com`), **path** (`/images/*`), header, method, or query → to a **target group**.
- **Target group** — a set of targets (EC2, IP, **Lambda**, containers) with its own health check.
- Features: **path/host-based routing**, **HTTP/2 & WebSockets**, **redirects & fixed responses**, **authentication** (Cognito/OIDC), **sticky sessions**, **TLS termination** (ACM certs).

```
Client → ALB listener :443
           ├─ rule: /api/*  → target group (api service)
           ├─ rule: /img/*  → target group (image service)
           └─ default       → target group (web)
```

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4. ⚡ NLB (Network Load Balancer)

Operates at **Layer 4** (TCP/UDP/TLS) for **extreme performance** and **low latency**.

- Handles **millions of requests/sec**; very low latency.
- Provides a **static IP per AZ** (and supports Elastic IPs) — useful for allow-lists/firewalls.
- **Preserves the client source IP** to targets.
- Can do **TLS termination**; great for non-HTTP protocols, gaming, IoT, and when you need a fixed IP.

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5. 🩺 Health Checks

- The LB periodically probes each target; only **healthy** targets receive traffic.
- Tunables: protocol/port/path, **interval**, **timeout**, **healthy/unhealthy thresholds**.
- A failing target is **drained** (existing connections finish via *deregistration delay*) and removed from rotation; recovery re-adds it.
- ELB health checks can also tell the **ASG** to replace an instance (see §10).

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6. 🧱 Key ELB Concepts

Concept	Meaning
Cross-zone load balancing	Evenly spreads traffic across targets in **all** AZs (default on for ALB; optional for NLB).
Sticky sessions	Pins a client to the same target via a cookie (ALB/CLB).
SSL/TLS termination	LB decrypts HTTPS using an **ACM** cert, forwarding HTTP to targets.
Connection draining (deregistration delay)	Lets in-flight requests finish before removing a target.
SNI	Multiple certs/domains on one ALB listener.
Access logs	Optionally log requests to S3.

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7. 📈 Auto Scaling Groups (ASG)

An **ASG** manages a fleet of EC2 instances between capacity bounds.

Setting	Role
Minimum	Never go below this many instances.
Desired	The target count right now.
Maximum	Never exceed this many.
Launch Template	Defines *what* to launch (AMI, type, SG, user data).
Subnets/AZs	Spread instances across multiple AZs for HA.

- The ASG keeps the running count at **Desired**, **replaces failed instances**, and registers/deregisters them with the ELB automatically.

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8. 📝 Launch Templates

- A **Launch Template** specifies the instance configuration the ASG uses: AMI, instance type, key pair, security groups, IAM role, user data, storage.
- **Versioned** (preferred over the older *launch configuration*) and supports **mixed instances** (combine types + On-Demand/Spot for cost).

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9. 🎯 Scaling Policies

Policy	How it scales
Target Tracking	Keep a metric at a target (e.g. CPU = 50%) — simplest, recommended.
Step Scaling	Add/remove N instances based on alarm magnitude.
Simple Scaling	One adjustment per alarm (older).
Scheduled Scaling	Change capacity on a schedule (e.g. scale up 9am).
Predictive Scaling	ML forecasts load and pre-scales ahead of demand.

- Policies are driven by **CloudWatch alarms** (CPU, request count per target, custom metrics).
- **Cooldown** periods prevent thrashing between rapid scale events.

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10. 🔄 Lifecycle & Health

- **Health check type: EC2** (instance status checks) or **ELB** (the load balancer's health check). Using **ELB health checks** lets the ASG replace instances the app — not just the OS — considers unhealthy.
- **Lifecycle hooks** pause instances in `Pending` / `Terminating` so you can run setup (bootstrap) or cleanup (drain, deregister) before they go live or away.
- **Termination policies** decide which instance to remove on scale-in (e.g. oldest launch template, closest to billing hour).

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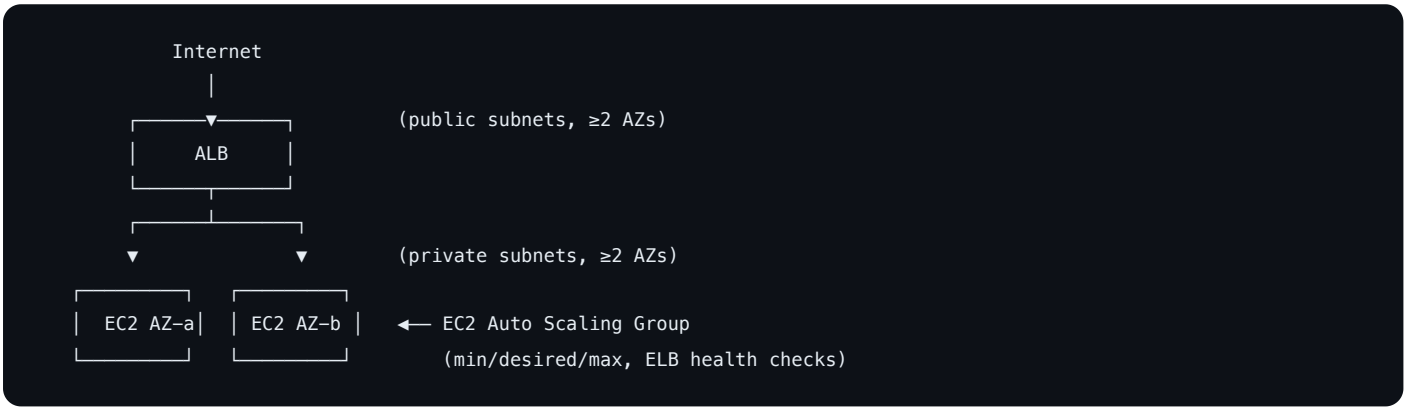
11. 🏛️ ELB + ASG: HA Architecture

The textbook highly-available web tier:



- ALB in **public** subnets across **≥2 AZs**; instances in **private** subnets across the same AZs.
- ASG keeps the fleet sized to load and **replaces unhealthy instances**; ELB only routes to healthy ones.
- Result: **elastic** (scales with demand) **+ self-healing** (failures auto-replaced) **+ multi-AZ** (survives an AZ outage).

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12. 💰 Pricing

- **ELB** — per **load-balancer-hour** + **LCU/NLCU** (capacity units measuring connections, bandwidth, rules processed).
- **Auto Scaling** — **free**; you only pay for the **EC2 instances** (and EBS) it launches.
- **Data transfer** out is charged as usual.

> 💸 **Cost tips:** Auto Scaling itself is free — scale in aggressively for non-prod · use **mixed instances + Spot** in the ASG for cheap capacity · right-size before scaling out · consolidate services behind one ALB (host/path routing) instead of many LBs.

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13. ⌨️ Common CLI Commands

```
# Load balancer (ELBv2 = ALB/NLB)
aws elbv2 create-load-balancer --name web-alb --subnets subnet-a subnet-b
aws elbv2 create-target-group --name web-tg --protocol HTTP --port 80 --vpc-id vpc-123
aws elbv2 create-listener --load-balancer-arn <arn> --protocol HTTP --port 80 \
  --default-actions Type=forward,TargetGroupArn=<tg-arn>

# Auto Scaling
aws autoscaling create-auto-scaling-group --auto-scaling-group-name web-asg \
  --launch-template LaunchTemplateName=web-lt --min-size 2 --max-size 6 \
  --desired-capacity 2 --vpc-zone-identifier "subnet-a,subnet-b" \
  --target-group-arns <tg-arn> --health-check-type ELB
aws autoscaling put-scaling-policy --auto-scaling-group-name web-asg \
  --policy-name cpu50 --policy-type TargetTrackingScaling \
  --target-tracking-configuration '{"PredefinedMetricSpecification":{"PredefinedMetricType":"ASGAverageCPUUtilization"},"
aws autoscaling describe-auto-scaling-groups
```

---

14. ✅ Best Practices

- Spread the **ELB and ASG across ≥2 AZs**; put instances in **private** subnets behind the LB.
- Use **ELB health checks** on the ASG so app-level failures trigger replacement.
- Prefer **Target Tracking** scaling (simple, robust); add **Scheduled/Predictive** for known patterns.
- Use **Launch Templates** + **mixed instances/Spot** for cost-efficient capacity.
- Set **cooldowns** to avoid thrashing; use **lifecycle hooks** for clean bootstrap/drain.
- Terminate gracefully with **connection draining** (deregistration delay).
- Consolidate microservices behind one **ALB** via host/path rules; use **NLB** for static IP / extreme performance.

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15. 🧠 Quick Mental Model

- **ELB = front door** distributing traffic to healthy targets across AZs; **ASG = engine room** scaling and healing the fleet.
- **ALB** (L7, host/path routing, web/microservices) · **NLB** (L4, ultra-fast, static IP) · **GWLB** (appliances) · CLB (legacy).
- ALB: **listener** → **rules** → **target groups** (each with a **health check**).
- **Health checks** keep traffic on healthy targets; **connection draining** finishes in-flight requests.
- ASG = **min / desired / max** + **launch template** across AZs; replaces unhealthy instances.
- **Scaling policies:** Target Tracking (default), Step, Scheduled, Predictive — driven by CloudWatch alarms.
- **ALB + ASG across ≥2 AZs = elastic + self-healing + highly available.** Auto Scaling is free; pay for instances.

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16. ❓ Common Interview Questions

Rapid-fire questions interviewers ask about ELB & Auto Scaling:

- **Q: ALB vs NLB?** — ALB = Layer 7 (HTTP/HTTPS), routes on host/path/headers, for web apps/microservices. NLB = Layer 4 (TCP/UDP), ultra-low latency, millions of req/s, static IP.
- **Q: How do you make a web app highly available?** — ALB + Auto Scaling Group across ≥2 AZs; instances in private subnets; ELB health checks for self-healing.
- **Q: What does an Auto Scaling Group give you?** — Elasticity (match capacity to demand) and resilience (auto-replace unhealthy/failed instances).
- **Q: EC2 vs ELB health check on the ASG?** — EC2 check = instance status only. ELB check = the load balancer's app-level check, so the ASG replaces instances the app considers unhealthy.
- **Q: What scaling policies exist?** — Target Tracking (keep a metric at a target), Step, Simple, Scheduled, Predictive.
- **Q: What is connection draining?** — Deregistration delay that lets in-flight requests finish before a target is removed.
- **Q: What is cross-zone load balancing?** — Distributes traffic evenly across targets in all AZs (default on for ALB).
- **Q: How does SSL termination work?** — The ALB holds an ACM cert, decrypts HTTPS, and forwards HTTP to targets (offloading TLS).
- **Q: Does Auto Scaling cost extra?** — No — it's free; you pay only for the EC2 instances and EBS it launches.
- **Q: How do you reduce capacity cost?** — Mixed instances with Spot in the ASG, right-sizing, and scheduled scale-in for non-prod.
